Nonlocal Means Regularized Sketched Reweighted Sparse and Low-Rank Subspace Clustering for Large Hyperspectral Images

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***Abstract*— Clustering is a common method for hyperspectral image (HSI) interpretation in the case of no labeled samples. Many subspace clustering methods have now been proposed for HSIs and have obtained remarkable success. However, because of the prohibitively large computational complexity induced by the self-dictionary representation, these methods suffer from the scalability issue and are ineffective for large HSIs. In this article, to address this issue, we focus on a scalable subspace clustering scheme and introduce the recently developed sketched subspace clustering (sketched-SC) model to HSI. The sketched-SC model is computationally inexpensive and is suitable for the large HSI clustering task as it constructs a compact yet expressive dictionary. However, several problems degrade the performance of sketched-SC, i.e., the inadequate mining of the structural information and no consideration of spatial information. In view of this, a novel scalable nonlocal means regularized sketched reweighted sparse and low-rank (NL-SSLR) SC algorithm is proposed for use with large HSIs. On the one hand, the SSLR representation model is constructed to explore the underlying local and global structural information of the HSIs at the same time. On the other hand, the nonlocal means regularization is used to fully explore the spatial correlation information and better account for the self-similarity of HSIs, to further boost the clustering performance. The experimental results obtained on two well-known hyperspectral data sets corroborate the superiority of the proposed algorithm over the other state-of-the-art HSI clustering methods.**

***Index Terms*— Clustering, large hyperspectral images (HSIs), nonlocal means regularization, sketched sparse subspace clustering (sketched-SSC), sparse and low-rank representation (SLRR).**

I. Introduction

HYPERSPECTRAL images (HSIs) can provide nearly continuous spectral information for an observed area, with the unique advantage of integrated and continuous spatial and spectral information [1]–[5]. Consequently, HSIs have very broad applications, including geological applications [6], atmospheric studies [7], hydrology studies [8], and military applications [9]. HSIs can allow us to better differentiate various materials through the diagnostic spectral differences. Clustering, which is also referred to as "unsupervised classification," is a basic technique in unsupervised HSI information extraction. As it has no need for labeled samples, clustering is more attractive than supervised classification in practical applications [10], [11]. In general, clustering is aimed at assigning similar pixels into groups based on certain similarity criteria. However, HSI clustering generally faces great challenges on account of the complex internal structure and serious spectral variability of HSIs [12], [13].

Up to now, a large amount of HSI clustering methods have been developed on the basis of different theories and working mechanisms [12]–[14]. Generally speaking, they can be roughly summed up to the following five models: 1) centroid-based ones; 2) density-based ones; 3) probability-based ones; 4) biology-based ones; and 5) graph-based ones [14].

The centroid-based methods iteratively optimize the centroids and the associated pixel assignments until the partition error meets the tolerance requirement, which are generally simple and efficient, with a wide utilization in practical applications. Examples of the centroid-based methods are k-means clustering [15], fuzzy c-means (FCM) clustering [16], adaptive FCM clustering with local information [17], and Albedo recovery FCM [18]. The density-based methods search for the partition patterns of pixels according to their local densities and relative distances in the feature space, which are relatively robust to the cluster shapes and noise. Typical examples of the density-based methods are clustering based on density peaks [19], clustering and active learning with nonlinear diffusion [20], spectral–spatial diffusion geometry clustering [21], and clustering based on nearest neighbor density [22]. The probability-based methods model the pixels based on a certain probability distribution function, which are commonly based on strict mathematical foundations and easily interpreted. The examples of the probability-based methods are the Gaussian mixture model [23], independent component analysis mixture model [24], and variational expectation maximization [25]. The biology-based methods segment HSIs based on biological models and principles, which can better simulate the complexity of HSIs with improved clustering performance. The typical

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examples of the biology-based methods are clustering based on the artificial immune network [26] and artificial bee colony with Markov random field [27]. The graph-based methods explore the underlying relationship between pixels using a similarity graph and partition the pixels by the use of a graph-cut method, which can flexibly model the relationship between pixels with a powerful information mining ability. Typical examples are spectral clustering [28], sequential spectral clustering [29], and clustering based on the minimax bridge [30]. However, most of these methods encounter some common bottlenecks, such as inaccurate modeling of the complex internal structure of the HSI and limited adaptation to the large spectral variability, which generally leads to an unsatisfactory clustering performance [12], [14]. Recently, a set of co-clustering methods have been proposed, which obtain a relatively good clustering effect by constructing the co-clustering framework, such as spectral–spatial co-clustering [31], fuzzy co-clustering with cluster tendency assessment [32], and fuzzy co-clustering with feature-reduction [33]. However, these methods are troubled by similar obstacles as above and generally have limited performance for HSIs.

Faced with these obstacles, subspace clustering (SC) approaches have been developed for use with HSIs, which are a relatively recent area of focus. The crux of these methods is to learn an informative and favorable adjacency matrix between pixels through self-dictionary learning [14], [34]. The typical representative method is sparse SC (SSC) [12], [35], which relieves the spectral variability by enforcing pixels of the same class with different spectra to lie in the same subspace. SSC then explores the latent adjacency between pixels through the sparse representation under the HSI data dictionary. Based on the SSC model, several improved versions have been put forward to further improve its practicability, such as multiobjective SSC (MOSSC) [36] and improved SSC for HSIs with missing pixels [37]. However, due to neglecting the spatial information, the clustering results of SSC are generally contaminated by serious salt-and-pepper noise. Therefore, many enhanced models integrating spatial information have been subsequently developed, such as ℓ_2-norm regularized SSC (L2-SSC) [38], kernel SSC with spatial max pooling (KSSC-SMP) [39], and high-order SSC with Markov random field (HSSC-MRF) [40]. Unfortunately, in spite of the virtues of the impressive clustering performance, these self-dictionary methods generally face a computational bottleneck due to the vast data volume of HSIs. In fact, for large HSIs, the large-scale sparse optimization and graph-cut problems are often insolvable, due to the memory limitations of computers, which greatly limits their scalability in practical applications.

Based on this fact, for the past few years, several scalable clustering methods have been developed specifically for use with large HSIs. Among the different methods, a representative type of method is the methods based on anchor graphs, such as fast spectral clustering with anchor graph (FSCAG) [41] and scalable graph clustering with nonnegative regularization (SGCNR) [13]. Although these methods are very efficient as they greatly reduce the complexity by the use of the anchor graph, they cannot perform as well as the SC methods because of their inaccuracy in learning the intrinsic adjacency between pixels.

In this article, in view of this, in the framework of SC, we propose a novel scalable nonlocal means regularized sketched reweighted sparse and low-rank (NL-SSLR) SC algorithm for use with large HSIs. The contributions of this article are fourfold.

First, the recently developed sketched-SC model is introduced to the large HSI clustering task [42]. Differing from SSC, sketched-SC explores the latent adjacency between pixels under a compact dictionary with much fewer atoms, to make the SC arrive at a computationally affordable level. However, as sketched-SC only takes the spectral information into account and cannot adequately mine the underlying structural information, directly applying it to HSIs commonly failed to obtain good results.

Second, with the aim of fully excavating the structural information of HSIs, an SSLR representation model is constructed to simultaneously explore the local and global structural information, to learn more favorable and discriminative representations. Therein, a reweighted sparse constraint is implemented to better promote the sparsity of the coefficient matrix and better explore the local structural information.

Third, from the point of view of the joint modeling of spectral–spatial information, the nonlocal means regularization is constructed to fully explore the abundant spatial correlation information of the HSI, to boost the self-similarity of the coefficient matrix, and to better enhance the smoothness of the cluster map.

Finally, the proposed NL-SSLR algorithm can effectively explore the intrinsic adjacency between pixels in a computationally tolerable way, which inherits the advantages of SC and broadens its applicability to a large degree, to make it feasible for use with large HSIs.

The structure of this article is as follows. The sketched-SC model is first reviewed in Section II, and then, the proposed NL-SSLR algorithm is described intimately in Section III. The experimental verification and the associated discussion are provided in Section IV. Section V summarizes the article and suggests several future research directions.

II. Sketched Subspace Clustering for Large Hyperspectral Images

The sketched-SC model is derived from the SSC model, with a similar clustering scheme. For the SSC-based approaches, the HSI data are employed as the dictionary [12], [38], i.e., $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N], \mathbf{Y} \in \mathbb{R}^{D\times N}$, with each column representing the associated pixel, and D and N denote the number of bands and the number of pixels, respectively. Each pixel is then promoted to be recovered by a few atoms from its own subspace in the following sparsity-promoting problem:

$$\min_{\mathbf{C}} \|\mathbf{C}\|_1 + \frac{\lambda}{2}\|\mathbf{Y} - \mathbf{Y}\mathbf{C}\|_F^2 + \alpha\Gamma(\mathbf{C}) \quad \text{s.t. } \operatorname{diag}(\mathbf{C}) = \mathbf{0} \tag{1}$$

where $\mathbf{C} \in \mathbb{R}^{N\times N}$ denotes the coefficient matrix, $\Gamma(\cdot)$ stands for certain spatial constraint, λ and α are two tradeoff parameters, and $\operatorname{diag}(\mathbf{C}) = \mathbf{0}$ is utilized to exclude the case of

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pixel self-representation. For convenience, SSC determines parameter λ as $\lambda = \gamma / \min_i \max_{j \neq i} \left| \mathbf{y}_i^T \mathbf{y}_j \right|$, where γ is an adjusted coefficient [12], [35]. Hence, λ is determined by γ in practice. The adjacency matrix $\tilde{\mathbf{W}}$ is generally constructed as $\tilde{\mathbf{W}} = |\mathbf{C}| + \left| \mathbf{C}^T \right|$, and then by employing spectral clustering to segment it, the final clustering results can be achieved [43], [44].

It should be noted that these self-dictionary models generally have high computational complexity and cannot deal with large HSIs. Specifically, the image dictionary-based representation leads to a large-sized representation coefficient matrix, which brings a heavy computational burden and large memory consumption for the subsequent spectral clustering segmentation, in addition to being computationally difficult for the sparse optimization problem in (1). Moreover, the complexity of these methods scales cubically with the number of pixels. When it comes to large HSIs, due to the very large data volume, the prohibitive computational complexity and memory requirements hinder the working of these self-dictionary SC methods.

To address the above problems, the sketched-SC model focuses on a scalable framework [42]. As is well known, hyperspectral pixels are commonly highly correlated, and thus, the original image dictionary is often highly redundant. For the sake of relieving the complexity caused by the high data volume of HSIs, sketched-SC commits to compress the original dictionary in a reasonable way, to construct a compact but expressive lightweight dictionary, i.e., $\mathbf{B} \in \mathbb{R}^{D \times n}$, where n denotes the number of atoms, and $n \ll N$. It then explores the latent adjacency between pixels under the sketched dictionary $\mathbf{B}$. On the premise of keeping the representation ability of the dictionary, sketched-SC pursues a sketched dictionary with as few atoms as possible, to reduce the computational complexity and memory consumption to an affordable level, while avoiding obvious performance degradation. By this means, sketched-SC can be applied to large HSIs.

For brevity, sketched-SC employs random projection to cut down the volume of the dictionary in the following way:

$$\mathbf{B} = \mathbf{YR} \tag{2}$$

where $\mathbf{R} \in \mathbb{R}^{N \times n}$ denotes the random projection matrix, which is utilized to compress $\mathbf{Y}$, and should preserve as much as possible of the effective information from $\mathbf{Y}$ to guarantee the expressive capability of $\mathbf{B}$. For this purpose, the noted Johnson–Lindenstrauss transform (JLT) matrix is utilized as the random projection matrix $\mathbf{R}$, given its good properties in information retention [42]. For a detailed introduction to JLT, we refer the reader to [42] and [45]. The construction process for the sketched dictionary is shown in Fig. 1.

With the sketched dictionary $\mathbf{B}$ utilized, the sketched-SC model can be formulated as follows:

$$\min_{\mathbf{A}} h(\mathbf{A}) + \frac{\lambda}{2} \|\mathbf{Y} - \mathbf{BA}\|_F^2 \tag{3}$$

where $\mathbf{A} \in \mathbb{R}^{n \times N}$ denotes the associated representation coefficient matrix, with a much smaller size than $\mathbf{C} \in \mathbb{R}^{N \times N}$, and $h(\cdot)$ denotes the structural prior constraint of the representation coefficient matrix $\mathbf{A}$. Similar to SSC, parameter λ is determined by γ in practice.

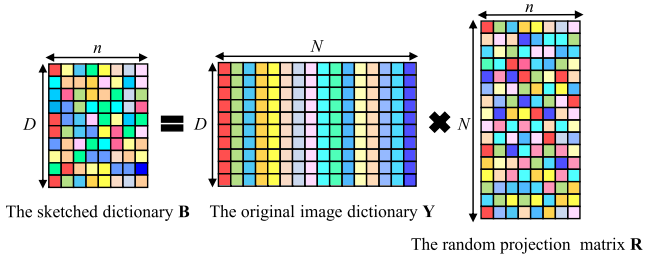


Fig. 1. Sketched dictionary construction.

Generally speaking, according to the choice of $h(\cdot)$, three specific models can be derived from (3), i.e., sketched sparse subspace clustering (Sketched-SSC), sketched low-rank representation (sketched-LRR), and sketched least-squares regression (sketched-LSR) [42].

For sketched-SSC, it can be modeled as:

$$\min_{\mathbf{A}} \|\mathbf{A}\|_1 + \frac{\lambda}{2} \|\mathbf{Y} - \mathbf{BA}\|_F^2 \tag{4}$$

which leads to a sparse representation coefficient matrix and is commonly solved with the least absolute shrinkage and selection operator (LASSO) method [46].

For sketched-LRR, it can be modeled as:

$$\min_{\mathbf{A}} \|\mathbf{A}\|_* + \frac{\lambda}{2} \|\mathbf{Y} - \mathbf{BA}\|_F^2 \tag{5}$$

which leads to an LRR coefficient matrix and is commonly solved with the singular value thresholding (SVT) method [47].

For sketched-LSR, it can be modeled as:

$$\min_{\mathbf{A}} \frac{1}{2} \|\mathbf{A}\|_F^2 + \frac{\lambda}{2} \|\mathbf{Y} - \mathbf{BA}\|_F^2 \tag{6}$$

which leads to a dense coefficient matrix and allows a closed-form solution as $\mathbf{A} = \lambda(\lambda \mathbf{B}^T \mathbf{B} + \mathbf{I})^{-1} \mathbf{B}^T \mathbf{Y}$.

Differing from SSC, it is impossible to construct the adjacency matrix $\tilde{\mathbf{W}}$ with $\mathbf{A}$ as $\tilde{\mathbf{W}} = |\mathbf{A}| + |\mathbf{A}^T|$ for spectral clustering since $\mathbf{A}$ is no longer a square matrix but has a size of $n \times N$. Therefore, sketched-SC constructs a k-nearest neighbor graph with the columns of $\mathbf{A}$, in the following two ways [42]:

$$\tilde{\mathbf{W}}_{i,j} = \begin{cases} 1, & \text{if } \mathbf{a}_j \in \mathrm{K}_i \text{ or } \mathbf{a}_i \in \mathrm{K}_j \\ 0, & \text{otherwise} \end{cases} \tag{7-1}$$

$$\tilde{\mathbf{W}}_{i,j} = \begin{cases} \exp\left(-\left\| \mathbf{a}_i - \mathbf{a}_j \right\|_2^2 / \sigma^2 \right), & \text{if } \mathbf{a}_j \in \mathrm{K}_i \text{ or } \mathbf{a}_i \in \mathrm{K}_j \\ 0, & \text{otherwise} \end{cases} \tag{7-2}$$

where $\mathbf{a}_i$ is the i th column of $\mathbf{A}$, and K_i is the ith neighbor, including the k columns closet to $\mathbf{a}_i$, according to the Euclidean distances. σ is a predefined constant, with $\sigma > 0$. In practice, (7-2) is generally adopted to construct the adjacency matrix. Then, by applying spectral clustering to $\tilde{\mathbf{W}}$, sketched-SC obtains its final clustering result [43], [44]. The sketched-SC model can be tabulated as shown in Algorithm 1.

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Algorithm 1 Sketched Subspace Clustering for Large HSIs

Input:
1) The HSI matrix $\mathbf{Y} = [\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_N], \mathbf{Y} \in \mathbb{R}^{D\times N}$;
2) The cluster number c and parameters γ and n.
Main body:
1) Construct the sketched dictionary $\mathbf{B}$ according to (2);
2) Obtain the coefficient matrix $\mathbf{A}$ from (4) or (5) or (6);
3) Create $\tilde{\mathbf{W}}$ with the columns of $\mathbf{A}$ according to (7-2);
4) Employ spectral clustering to segment $\tilde{\mathbf{W}}$.
Output:
Cluster map.

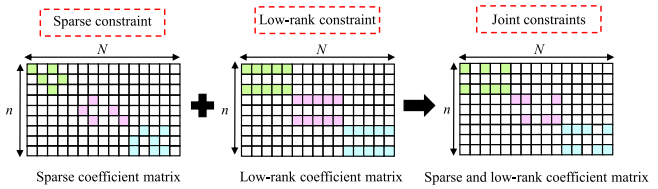


Fig. 2. SSLR representation model.

III. Nonlocal Means Regularized Sketched Reweighted Sparse and Low-Rank Subspace Clustering

On the basis of the sketched-SC framework, a new NL-SSLR algorithm is proposed for use with large HSIs. The proposed algorithm simultaneously learns the local and global structural information of the HSI through a sparse and low-rank joint constraint, and it fully explores the self-similarity and spatial autocorrelation information of the HSI by the nonlocal means regularization, to further improve the clustering performance.

A. Sketched Reweighted Sparse and Low-Rank Subspace Clustering for HSIs

Many previous studies have shown that, through individually recovering a pixel with only a few atoms, the sparsity regularization can learn the local structural information of the HSIs, and by minimizing the rank of the representation coefficient matrix under joint representation with all the atoms, the low-rank regularization can effectively capture the global structural information of the HSIs [48]–[50]. However, the sketched-SC model considers one of these two constraints in isolation, as in (4) or (5), and it neglects the relationship and complementation between them, which leads to inadequate mining of the structural information, and limits the final clustering accuracy to some extent.

On this account, the SSLR model is constructed to fully excavate the underlying structural information of the HSIs, as shown in Fig. 2. To be specific, the SSLR model is constructed as:

$$\min_{\mathbf{A}} \|\mathbf{A}\|_1 + \beta\|\mathbf{A}\|_* + \frac{\lambda}{2}\|\mathbf{Y} - \mathbf{B}\mathbf{A}\|_F^2 \tag{8}$$

where β is a regularization parameter, which is used to tradeoff the sparsity and the rank-lank property of $\mathbf{A}$. As the ℓ_1-norm regularization cannot induce an ideal sparse solution, the reweighted ℓ_1-norm regularization is utilized to better explore the local structural property of the HSIs [51], [52]. In this way, (8) can be transformed as:

$$\min_{\mathbf{A},\mathbf{W}} \|\mathbf{W} \odot \mathbf{A}\|_1 + \beta\|\mathbf{A}\|_* + \frac{\lambda}{2}\|\mathbf{Y} - \mathbf{B}\mathbf{A}\|_F^2 \tag{9}$$

where $\mathbf{W}$ is a learned weight matrix, as defined in (10), which can be updated in the iterative process [52]. $\odot$ is the Kronecker product and denotes the element-wise multiplication operation.

$$\mathbf{W}_{i,j} = \frac{e_2}{\mathbf{A}_{i,j} + e_1} \tag{10}$$

where $\mathbf{W}$ is set as the derivative of $\mathbf{A}$, which can better promote the sparsity by punishing the larger elements. e_1 and e_2 are two small constants to avoid overweighting, which are both set to 0.001 in practice.

Through sparse and low-rank joint modeling, as shown in (9), the underlying structural information and partition pattern of the pixel set can be more effectively learned, which can further enhance the discriminability of the coefficient features and can help each pixel to be segmented into the correct subspace.

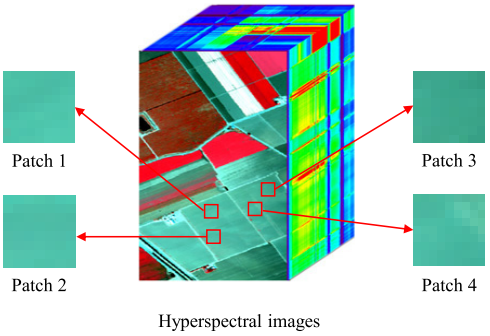


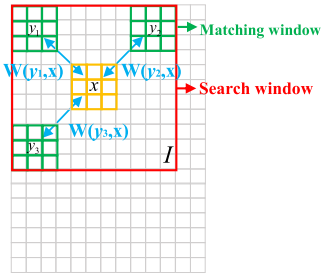
Fig. 3. Spatial self-similarity of hyperspectral imagery.

B. Nonlocal Means Regularized SSLR for HSI Clustering

As is well known, HSIs have strong spatial self-similarity and contain abundant spatial autocorrelation information [53]–[56], including rich nonlocal correlation information, as well as local neighborhood information, as shown in Fig. 3. Many studies have shown that incorporating the spatial correlation information can provide more support for HSI interpretation [12], [14], [17]. However, the traditional SC algorithms only consider the local contextual information in the four- or eight-neighborhood and treat the contribution of each pixel equally in the field. Therefore, most of the abundant spatial correlation information in the whole image is wasted, and the differences between pixels are ignored, which leads to insufficient and inaccurate spatial information exploration, and would greatly degrade the final clustering accuracy.

Based on this fact, it is natural to incorporate the nonlocal spatial correlation into clustering analysis, to better explore the spatial similarity of the HSI, and further improve the clustering performance. The nonlocal means filter has been proven to be effective in extracting the nonlocal correlation information of the image [57] and has been successfully

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The 2D map of each row of **A**

Fig. 4. Principle of the nonlocal means filter.

applied in HSI denoising, classification, and unmixing tasks [58]–[60]. Therefore, to fully excavate the spatial correlation information of the HSI, an effective nonlocal means regularization approach is proposed, based on the nonlocal means filter, to incorporate the nonlocal correlation information in the self-expression model, to better restrain the various singular point mutations in the plot and further promote the smoothness of the clustering result.

To be specific, the nonlocal means filter, which can be defined as shown in (11), with the principle shown in Fig. 4, is first employed on the 2-D map of each row of the coefficient matrix **A**, according to the working mechanism of self-representation [12], [14], [38]. Next, through reshaping the filtered 2-D maps into 1-D row coefficients, the nonlocal coefficient matrix $\bar{\mathbf{A}}_{NL} \in \mathbb{R}^{n\times N}$ can be obtained.

$$\text{NLfilter}(x) = \sum_{y\in I} w(x,\, y) \times \text{block}(y) \tag{11}$$

where x and y denote the matching window (or the dependent neighborhood), at a size of $s \times s$, in the search window I, at a size of $S \times S$, where block(y) refers to the pixels in y, and $w(x, y)$ represents the corresponding weight, which can be calculated based on the heat kernel distance between the two matching windows, with the kernel parameter set as 10 in practice.

Then, based on the nonlocal coefficient matrix $\bar{\mathbf{A}}_{NL}$, the nonlocal means regularization can be constructed as the following formulation:

$$\text{NLM}(\mathbf{A}) = \left\|\mathbf{A} - \bar{\mathbf{A}}_{\text{NL}}\right\|_F^2. \tag{12}$$

By fully excavating the spatial correlation information of the HSI, the nonlocal means regularization can boost the self-similarity of **A** and better enhance the smoothness of the cluster map. Differing from the local spatial constraints, the nonlocal means regularization utilizes all the pixels in the search window to calculate the spatial correlation information. As a result, it can better excavate the spatial information of the HSI. In addition, the nonlocal means regularization comprehensively considers the geometrical configuration in the whole neighborhood, rather than a single pixel, and adaptively calculates the relative contributions of the neighboring pixels, according to their similarities with the central pixel. Hence, it can more accurately describe the central pixel.

Finally, with the nonlocal means regularization integrated into the SSLR model, the proposed NL-SSLR algorithm can be modeled as:

$$\min_{\mathbf{A},\mathbf{W}} \|\mathbf{W}\odot\mathbf{A}\|_1 + \beta\|\mathbf{A}\|_* + \frac{\lambda}{2}\|\mathbf{Y}-\mathbf{B}\mathbf{A}\|_F^2 + \frac{\alpha}{2}\left\|\mathbf{A}-\bar{\mathbf{A}}_{NL}\right\|_F^2 \tag{13}$$

where α is the regularization parameter, which trades off the spectral term and spatial term.

C. Optimization of the NL-SSLR Algorithm

The widely used alternating direction method of multipliers (ADMMs) method is employed to solve the proposed NL-SSLR algorithm [61], with the optimization process briefly described as follows.

First, two auxiliary variables for coefficient matrix **A**, i.e., **Z** and **C**, are introduced into (13) to separate the variables:

$$\begin{aligned}\min_{\mathbf{A},\mathbf{Z},\mathbf{C},\mathbf{W}} &\|\mathbf{W}\odot\mathbf{Z}\|_1 + \beta\|\mathbf{C}\|_* \\ &+\frac{\lambda}{2}\|\mathbf{Y}-\mathbf{B}\mathbf{A}\|_F^2 + \frac{\alpha}{2}\left\|\mathbf{A}-\bar{\mathbf{A}}_{NL}\right\|_F^2 \\ \text{s.t. } &\mathbf{A}=\mathbf{Z}, \mathbf{A}=\mathbf{C}.\end{aligned} \tag{14}$$

Second, by introducing two Lagrange multipliers, i.e., $\mathbf{D} \in \mathbb{R}^{n\times N}$ and $\mathbf{E} \in \mathbb{R}^{n\times N}$, the augmented Lagrange penalty function is constructed as shown in the following equation:

$$\begin{aligned}\min_{\mathbf{A},\mathbf{Z},\mathbf{C},\mathbf{W},\mathbf{D},\mathbf{E}} &\|\mathbf{W}\odot\mathbf{Z}\|_1 + \beta\|\mathbf{C}\|_* + \frac{\lambda}{2}\|\mathbf{Y}-\mathbf{B}\mathbf{A}\|_F^2 \\ &+\frac{\alpha}{2}\left\|\mathbf{A}-\bar{\mathbf{A}}_{NL}\right\|_F^2 + \frac{\rho}{2}\|\mathbf{Z}-\mathbf{A}-\mathbf{D}/\rho\|_F^2 \\ &+\frac{\rho}{2}\|\mathbf{C}-\mathbf{A}-\mathbf{E}/\rho\|_F^2\end{aligned} \tag{15}$$

where ρ is the penalty parameter, which is set as 1000 in practice.

Equation (15) can then be divided into four simple subproblems, to update each variable.

For **A**, it can be updated as follows:

$$\begin{aligned}&\mathbf{A}^{(k+1)} \\ &= \left(\lambda\mathbf{B}^T\mathbf{B} + \alpha\mathbf{I} + \rho\mathbf{I} + \rho\mathbf{I}\right)^{-1} \\ &\quad\times\left(\lambda\mathbf{B}^T\mathbf{Y} + \alpha\bar{\mathbf{A}}_{NL}^{(k)} + \rho\mathbf{Z}^{(k)} - \mathbf{D}^{(k)} + \rho\mathbf{C}^{(k)} - \mathbf{E}^{(k)}\right).\end{aligned} \tag{16}$$

The nonlocal coefficient $\bar{\mathbf{A}}_{NL}$ can then be updated with $\mathbf{A}^{(k+1)}$, as shown in (11).

For **Z**, it is updated according to (17), with the LASSO method [46]:

$$\min_{\mathbf{Z}}\left\|\mathbf{W}^{(k)}\odot\mathbf{Z}\right\|_1 + \frac{\rho}{2}\left\|\mathbf{Z}-\mathrm{A}^{(k+1)}-\mathbf{D}^{(k)}\right\|_F^2. \tag{17}$$

The weight matrix **W** can then be updated with $\mathbf{Z}^{(k+1)}$, as shown in (10).

For **C**, it is updated according to the following equation, with the SVT method [47]:

$$\min_{\mathbf{C}} \beta\|\mathbf{C}\|_* + \frac{\rho}{2}\left\|\mathbf{C}-\mathbf{A}^{(k+1)}-\mathbf{E}^{(k)}\right\|_F^2. \tag{18}$$

Finally, **D** and **E** are updated with $\mathbf{A}^{(k+1)}$, $\mathbf{Z}^{(k+1)}$, and $\mathbf{C}^{(k+1)}$:

$$\mathbf{D}^{(k+1)} = \mathbf{D}^{(k)} + \rho\left(\mathbf{A}^{(k+1)} - \mathbf{Z}^{(k+1)}\right) \tag{19-1}$$

$$\mathbf{E}^{(k+1)} = \mathbf{E}^{(k)} + \rho\left(\mathbf{A}^{(k+1)} - \mathbf{C}^{(k+1)}\right). \tag{19-2}$$

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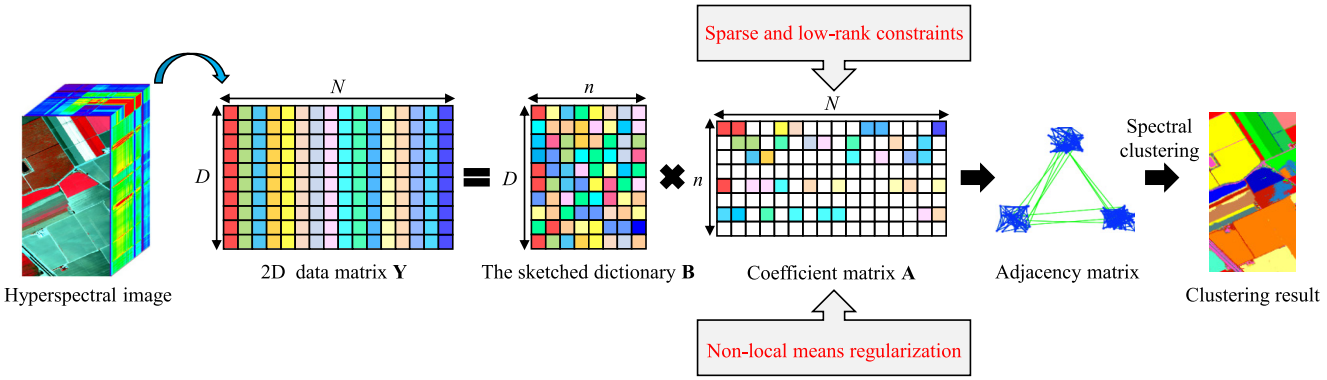


Fig. 5. Working scheme of the proposed NL-SSLR algorithm.

Algorithm 2 Non-Local Means Regularized Sketched Reweighted Sparse and Low-Rank Subspace clustering for Large HSIs

Input:
1) The HSI matrix $\mathbf{Y} = [\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_N], \mathbf{Y} \in \mathbb{R}^{D \times N}$;
2) The cluster number c and parameters γ, α, β, n, S,and s.
Main body:
1) Construct the sketched dictionary $\mathbf{B}$ according to (2);
2) Solve (13) by means of ADMM to obtain the coefficient matrix $\mathbf{A}$, looping until convergence or $k > 100$:
   1. Update $\mathbf{A}$ according to (16);
   2. Update $\mathbf{Z}$ according to (17) with the LASSO method;
   3. Update $\mathbf{C}$ according to (18) with the SVT method;
   4. Update $\mathbf{D}$ and $\mathbf{E}$ according to (19);
   5. $k = k + 1$;
3) Create $\tilde{\mathbf{W}}$ with the columns of $\mathbf{A}$ according to (7-2);
4) Employ spectral clustering to segment $\tilde{\mathbf{W}}$.
Output:
Cluster map.

Repeating the above optimization process until convergence, i.e., $\|\mathbf{A}^{(k)} - \mathbf{Z}^{(k)}\|_F^2 / \|\mathbf{A}^{(k)}\|_F^2 \leq \varepsilon^{\text{pri}}$, $\|\mathbf{A}^{(k)} - \mathbf{C}^{(k)}\|_F^2 / \|\mathbf{A}^{(k)}\|_F^2 \leq \varepsilon^{\text{pri}}$, and $\|\mathbf{A}^{(k)} - \mathbf{A}^{(k-1)}\|_F^2 / \|\mathbf{A}^{(k)}\|_F^2 \leq \varepsilon^{\text{dual}}$, or the iteration times exceeding the predefined maximum value, i.e., 100, where ε^{pri} and $\varepsilon^{\text{dual}}$ are the two predefined threshold values.

Similar to sketched-SC, with the obtained $\mathbf{A}$, $\tilde{\mathbf{W}}$ can be constructed as in (7-2). Then, by applying spectral clustering to it, NL-SSLR obtains its final clustering result [43], [44]. In general, the proposed NL-SSLR algorithm can be tabulated as in Algorithm 2, with the working scheme shown in Fig. 5.

IV. Experiment and Discussion

A. Experimental Settings

The proposed NL-SSLR algorithm was thoroughly tested using two widely used HSIs, i.e., the Indian Pines image and the Salinas image, with eight benchmark methods: FCM [16], FCM-S1 [62], SSC [12], [35], L2-SSC [38], scalable spectral clustering with cosine similarity (scalable-SC) [63], FSCAG [41], SGCNR [13], and sketched-SSC [42], with the last four algorithms being state-of-the-art large HSI clustering methods. Both images contain many subtypes of vegetation, which are difficult to distinguish. The first image is internationally recognized as being particularly challenging data for HSI interpretation, and the second image is of a large size, which is very challenging for SC.

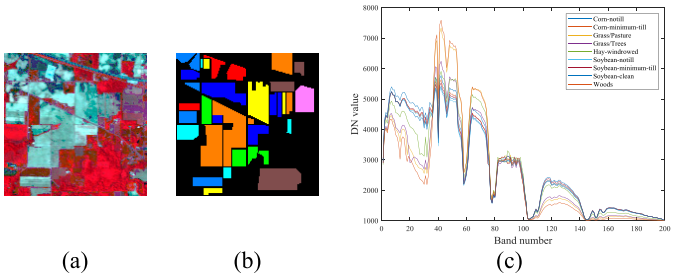


Fig. 6. Indian Pines experiment. (a) Original image (R: 40, G: 30, B: 20). (b) Ground truth. (c) Mean spectra of the nine classes.

As in most clustering methods, the number of clusters was entered manually, and the thematic information of each cluster was artificially determined through referencing with the original image. For each clustering method, the parameters were tuned to be optimal. For SGCNR and FSCAG, the recommended parameters were utilized [13], [41]. The results are the average of five independent Monte Carlo experiments. The clustering results are evaluated both qualitatively and quantitatively. The producer's accuracy, user's accuracy, overall accuracy (OA), kappa, purity index [64], and run time are provided for each experiment.

B. Indian Pines Image Experiment

The Indian Pines image was selected for the first experiment, which was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Northwestern Indiana, U.S., in 1992. This image has a 20-m spatial resolution and is of 145 × 145 pixels in size, with 200 bands retained for the spectral analysis after removing 20 bad bands. This scene includes 16 classes in total. Therein, nine main classes were utilized for the experimental analysis, as in [14] and [65]. Due to the high similarity of the classes, as shown in Fig. 6(c), this image is very challenging for

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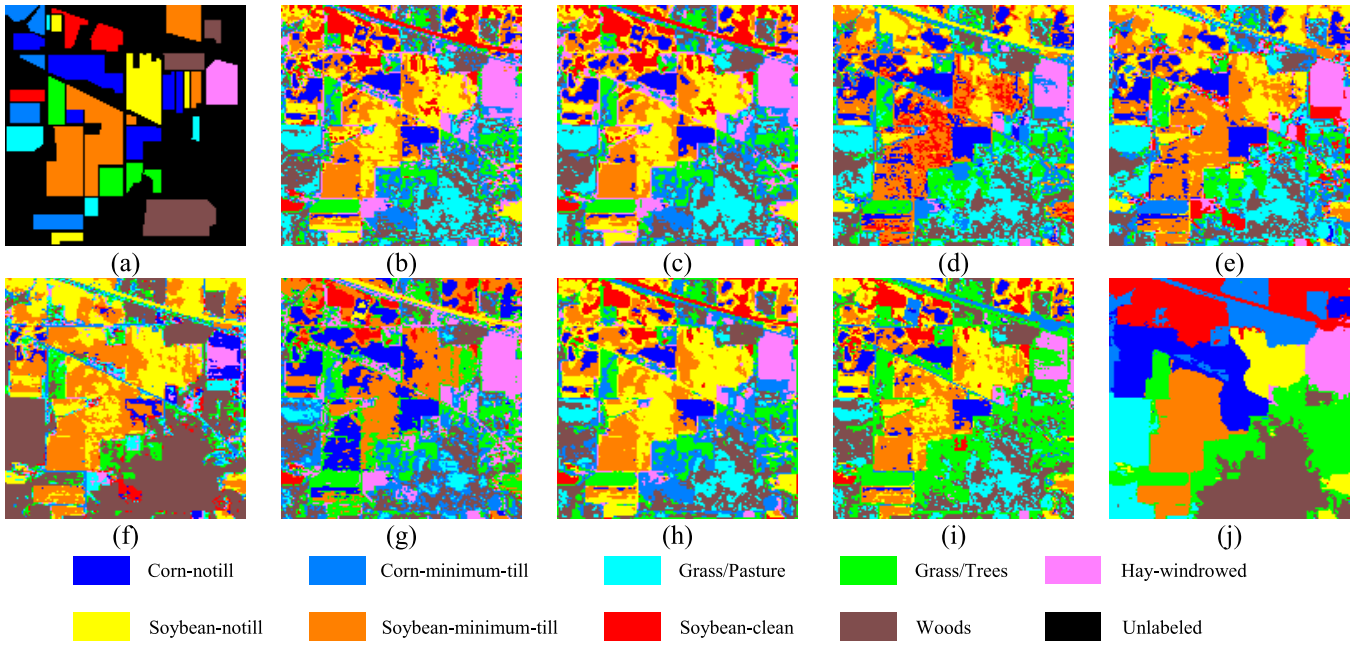


Fig. 7. Cluster maps of the Indian Pines experiment. (a) Ground truth. (b) FCM. (c) FCM-S1. (d) SSC. (e) L2-SSC. (f) Scalable-SC. (g) SGCNR. (h) FSCAG. (i) Sketched-SSC. (j) NL-SSLR.

TABLE I
PARAMETER SETTINGS FOR THE INDIAN PINES EXPERIMENT

Method	Parameters
FCM	$c=9$, $e=2$
FCM-S1	$c=9$, $e=2$, $\alpha=0.7$
SSC	$c=9$, $\gamma=400$
L2-SSC	$c=9$, $\gamma=200$, $\alpha=1\times10^3$
Scalable-SC	$c=9$, $\eta=0.5$
FSCAG	$c=9$, $m=500$, $k=5$, $\alpha=0.6$
SGCNR	c=9, $m=500$, $k=5$, ρ=1.1, μ=0.1
Sketched-SSC	$c=9$, $\gamma=600$, n=200
NL-SSLR	$c=9$, $\gamma=800$, n=100, $\beta=1$, $\alpha=3$, $S=5$, $s=2$

c : the number of clusters; e : fuzzy exponential; n : the number of atoms; η : the fraction of outliers to be removed; γ, α, β : regularization parameters; μ, ρ : penalty parameters; S : the search window size; s : the matching window size.

clustering. In Fig. 6, the original image and the ground truth are also provided. The parameter settings for this experiment are shown in Table I.

The experimental results are given in Fig. 7 and Table II, which clearly corroborate that spatial correlation information can help to improve the clustering performance. In addition, the proposed NL-SSLR algorithm performs much better compared with the other methods.

In particular, FCM fails to distinguish different land-cover classes in this scene, with a large number of misclassifications. For instance, there is a lot of confusion between Soybean-no-till and Soybean-minimum-till. FCM obtains a relatively good recognition effect for the Hay-windrowed class, with a producer's accuracy of 98.95% and a user's accuracy of 69.56%. However, for many other classes, the precisions are under 50%. Consequently, the OA of FCM is at a low level, i.e., 43.03%. By integrating the spatial information to help with the spectral analysis, FCM-S1 improves the clustering accuracy and smooths the noise, to a certain degree. It can also be seen that, for most classes, the producer's accuracy has been improved, to some extent.

Compared with these two traditional methods, SSC exhibits a great potential in HSI clustering owing to the superiority of subspace in the HSI modeling, and it successfully distinguishes most classes. For the Grass/Trees and Woods classes, SSC obtains much higher producer's accuracy and user's accuracy, i.e., 67.67% and 60.16% and 96.67% and 96.09%, compared with FCM and FCM-S1. However, due to ignoring the abundant spatial information of the HSI, there is still a lot of noise and misclassifications in the cluster map of SSC. Therefore, the OA of SSC is only 45.48%. In comparison, L2-SSC significantly improves the clustering accuracy by incorporating the spatial information to enhance the discrimination and smooth the within-class noise. Taking Grass/Trees and Soybean-minimum-till as examples, the producer's accuracy is improved from 67.67% and 48.92% to 76.71% and 67.09%, with the user's accuracy improved from 96.67% and 45.03% to 98.94% and 51.78%, respectively. Overall, L2-SSC obtains the second-best clustering accuracy in this scene, with an OA of 51.01%, which demonstrates its effectiveness and superiority. However, it should be noted that, because of the large computational complexity of SC, both SSC and L2-SSC are at a very large time cost, which greatly degrades their scalability in practice.

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TABLE II
QUANTITATIVE ASSESSMENT OF THE INDIAN PINES EXPERIMENT

Method	Class	FCM	FCM-S1	SSC	L2-SSC	Scalable-SC	SGCNR	FSCAG	Sketched-SSC	NL-SSLR
Producer's accuracy (%)	1	35.78	37.90	**48.39**	44.33	7.63	39.27	28.60	33.52	48.00
	2	0	0	0.24	0.24	1.69	8.58	3.01	**9.49**	0
	3	56.73	60.25	59.42	**65.01**	19.46	39.09	51.76	34.87	52.67
	4	52.19	54.79	67.67	76.71	61.92	59.12	63.42	76.27	**83.10**
	5	98.95	**100**	88.08	**100**	47.49	90.79	99.71	91.30	**100**
	6	49.38	49.59	34.36	45.68	59.67	27.86	60.39	49.22	**70.91**
	7	42.93	42.62	48.92	67.09	58.70	50.26	38.12	47.12	**73.43**
	8	33.73	34.57	1.52	0	1.35	15.41	28.20	25.63	**75.04**
	9	47.43	45.68	60.16	49.96	**99.76**	48.08	58.04	67.42	70.97
User's accuracy (%)	1	52.95	53.34	45.22	48.62	26.39	44.81	**65.96**	59.34	55.23
	2	0	0	0.48	1.30	**35.00**	3.64	3.14	6.67	0
	3	30.34	30.04	34.41	34.97	49.74	36.78	33.73	19.62	**80.00**
	4	94.54	93.90	96.67	**98.94**	97.20	66.99	79.23	72.38	72.56
	5	69.56	68.92	98.14	93.91	69.63	62.64	94.51	87.86	**99.63**
	6	20.78	21.01	28.64	23.97	21.45	15.98	21.08	24.32	**55.98**
	7	51.92	52.89	45.03	51.78	43.89	45.90	48.35	51.22	**78.71**
	8	21.93	22.76	1.01	0	12.90	19.95	22.10	24.35	**30.22**
	9	88.76	91.02	96.09	**98.29**	72.03	77.11	92.73	86.24	85.58
OA (%)		43.03	43.55	45.48	51.01	45.34	42.14	43.83	47.21	**63.45**
Kappa		0.3427	0.3497	0.3593	0.4173	0.3447	0.3198	0.3530	0.3861	**0.5741**
Purity		0.5528	0.5596	0.5489	0.5647	0.5120	0.4843	0.5642	0.5499	**0.6594**
Time (s)		129	381	32764	13532	**1.25**	7.42	1.44	78	9192

1: Corn-notill; 2: Corn-minimum-till; 3: Grass/Pasture; 4: Grass/Trees; 5: Hay-windrowed; 6: Soybean-notill; 7: Soybean-minimum-till; 8: Soybean-clean; 9: Woods

On the contrary, the three scalable clustering methods, i.e., Scalable-SC, SGCNR, and FSCAG, are very efficient and have very low time costs. However, due to their weakness in pixel adjacency mining and their lack of adaptability to the large spectral variability of HSIs, all three methods fail to obtain a good performance. The scalable-SC, SGCNR, and FSCAG methods obtain relatively low OAs of only 45.34%, 42.14%, and 43.83%. Specifically, SGCNR performs poorly in this scene, with serious misclassifications. Comparatively speaking, FSCAG effectively enhances the clustering performance by considering the spatial neighborhood information. Taking the Grass/Trees class as an example, the producer's accuracy and user's accuracy are improved from 59.12% and 66.99% to 63.42% and 79.23%, respectively, which further indicates the importance of spatial correlation information. By better modeling the similarity between pixels, scalable-SC performs better in this scene, and for the Woods class, it obtains the highest producer's accuracy.

By comparison, the two sketched SC algorithms perform much better in this experiment. Relative to SSC, with the help of the compact yet expressive dictionary, sketched-SSC significantly reduces the time cost. The run time is reduced from 32764 to only 78 s, which demonstrates the superiority of the sketched model. In addition, due to inheriting the advantages of SC, sketched-SSC effectively distinguishes most of the classes and obtains a comparable clustering result to SSC. However, because of incompletely mining the structural information and ignoring spatial information, sketched-SSC performs limitedly to obtain a clustering result with serious misclassifications, which is far from satisfactory. Compared with sketched-SSC, the proposed NL-SSLR algorithm greatly improves the clustering accuracy and obtains a much smoother cluster map. The reasons can be that NL-SSLR simultaneously excavates the local and global structural information through the sparse and low-rank joint representation to learn more favorable and discriminant features to reduce misclassifications and fully explores the abundant spatial correlation information through the nonlocal means regularization to boost the self-similarity of the coefficient matrix and smooth the within-class noise to enhance the homogeneity of the cluster map. For instance, the producer's accuracy for the Soybean-minimum-till and Soybean-clean classes is improved from 47.12% and 25.63% to 73.43% and 75.04%, with the user's accuracy improved from 51.22% and 24.35% to 78.71% and 30.22%, respectively. In addition, NL-SSLR achieves the highest or the second-highest precisions for most classes. Overall, NL-SSLR obtains the best cluster map, with the highest clustering accuracy, i.e., an OA of 63.65% and a kappa of 0.5741. Compared with sketched-SSC, NL-SSLR achieves an improvement in OA of over 16% and exhibits a much higher accuracy than the other benchmark methods.

Furthermore, NL-SSLR also achieves the optimal purity index in this scene, i.e., 0.6594. This indicates that NL-SSLR can promote the spatial homogeneity and smoothness of the cluster map, which further suggests its effectiveness. On the other hand, as the nonlocal means regularization is relatively

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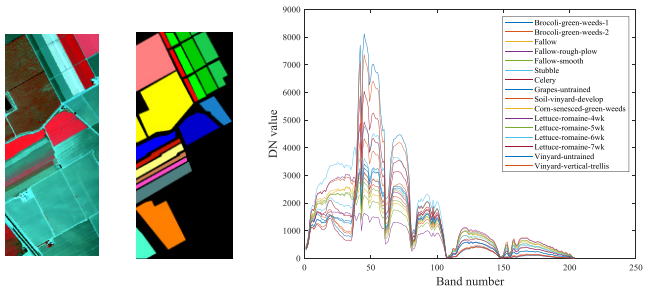


Fig. 8. Salinas experiment. (a) Original image (R: 50, G: 27, B: 17). (b) Ground truth. (c) Mean spectra of the 16 classes.

TABLE III
PARAMETER SETTINGS FOR THE SALINAS EXPERIMENT

Method	Parameters
FCM	$c=16$, $e=2$
FCM-S1	$c=16$, $e=2$, $\alpha=0.8$
SSC	---
L2-SSC	---
Scalable-SC	$c=16$, $\eta=0.05$
FSCAG	$c=16$, $m=1000$, $k=5$, $\alpha=0.8$
SGCNR	c=16 , $m=500$, $k=5$, ρ=1.1 , μ=0.1
Sketched-SSC	$c=16$, $\gamma=800$, n=150
NL-SSLR	$c=16$, $\gamma=800$, n=150 , $\beta=0.1$, $\alpha=0.01$, $S=11$, $s=2$

c : the number of clusters; e : fuzzy exponential; n : the number of atoms; η : the fraction of outliers to be removed; γ , α , β : regularization parameters; μ , ρ : penalty parameters; S : the search window size; s : the matching window size.

time-consuming, the time cost of NL-SSLR is relatively high. However, relative to SSC and L2-SSC, due to the sketched scheme, the run time is greatly reduced. In addition, with the help of advanced computer and high-performance computing technology, this problem will be well solved.

C. Salinas Image Experiment

The AVIRIS Salinas image was utilized for the second experiment, which covers an agricultural area in Salinas Valley, U.S. Through removing 20 bands of poor quality, there are 204 spectral measurements retained. The spatial resolution of the image is 3.7 m. The image has a much larger size of 512×217 pixels, with 111 104 pixels in total, which is very challenging for SC. As in the first experimental image, this image includes many subtypes of vegetation, with 16 classes in total. Many classes are very similar to each other, as shown in Fig. 8, which leads to large difficulties for clustering. The parameter settings for this experiment are shown in Table III.

The clustering results are shown in Fig. 9 and Table IV. From the experimental results, it can be noticed that, due to the high computational complexity, the two SC methods, i.e., SSC and L2-SSC, fail to cluster this large image. With the help of the sketched scheme, sketched-SSC and NL-SSLR show good scalability and can work well in this large scene while inheriting the advantages of SC. Generally speaking, the spectral–spatial models again obtain much better clustering results, and the proposed NL-SSLR algorithm again exhibits a large superiority over the other methods in this experiment.

To be specific, FCM again yields a bad clustering result in this experiment, with a lot of misclassifications and within-class noise. FCM obtains a fragmented cluster map, with an OA of only 53.51% and a purity value of only 0.6660. In comparison, with the help of the spatial information, FCM-S1 performs much better to obtain a much higher OA of 62.61%. It effectively smooths the within-class noise to obtain a much smoother cluster map. However, due to the drawbacks of the centroid-based models, FCM-S1 still shows a limited performance, with a lot of misclassifications. Comparatively speaking, the three scalable spectral clustering methods perform much better in this scene, with higher clustering precisions and much lower time costs. Overall, these three methods obtain very similar accuracies in this scene. Specifically, scalable-SC effectively recognizes most of the classes, with a run time of only 24 s. For the Corn-senesced-green-weeds class, it achieves the second-best producer's accuracy and user's accuracy in this experiment, i.e., 81.78% and 86.98%, respectively. By incorporating the spatial information, FSCAG performs slightly better than SGCNR and obtains an improved OA of 64.95%. It also has the lowest time cost of only 13 s. However, due to the limited ability to mine the pixel adjacency, it fails to obtain satisfactory clustering results in this scene.

Compared with the three scalable methods, sketched-SSC exhibits its superiority and obtains a much higher OA of 71.17%. This is achieved by more effectively modeling the large spectral variability of the HSI with subspaces and more accurately exploring the relationship between pixels through the sparse representation. In general, sketched-SSC obtains the best or second-best precisions for many classes. Compared with sketched-SSC, NL-SSLR greatly improves the clustering performance by better excavating the underlying structural information of the HSIs with the joint constraints and further promoting the homogeneity of the cluster map with the nonlocal means regularization, achieving the best clustering accuracy, i.e., the best OA of 77.43% and the best kappa of 0.7455. It obtains an improvement in OA of over 6%, compared with sketched-SSC, and shows an improvement of over 12%, compared with the other benchmarks. NL-SSLR effectively recognizes most of the classes, with higher precision. For Celery and Vinyard-vertical-trellis, both the producer's accuracy and user's accuracy are over 98%, which are superior to the precisions of the other methods. Furthermore, NL-SSLR effectively reduces the misclassifications and smooths the within-class noise, to obtain the best purity value of 0.7786. However, due to the high computational complex of the nonlocal means regularization and the larger size of the search window in this scene, the time cost of NL-SSLR is very high, which is a defect to be overcome in the future work.

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TABLE IV
QUANTITATIVE ASSESSMENT OF THE SALINAS EXPERIMENT ("OM" MEANS "OUT OF MEMORY")

Method	Class	FCM	FCM-S1	SSC	L2-SSC	Scalable-SC	SGCNR	FSCAG	Sketched-SSC	NL-SSLR
Producer's accuracy (%)	1	**99.75**	99.68	OM	OM	0	73.93	57.93	39.86	99.65
	2	39.32	50.05	---	---	98.12	80.37	96.42	98.60	**99.01**
	3	**56.22**	35.32	---	---	17.46	44.86	53.16	9.35	32.48
	4	**99.24**	98.22	---	---	16.66	69.73	58.06	59.68	79.64
	5	91.04	84.97	---	---	77.89	83.86	90.46	**99.16**	99.13
	6	86.29	95.36	---	---	99.72	86.69	83.86	**99.76**	99.50
	7	90.49	83.30	---	---	49.82	80.60	44.72	99.58	**99.90**
	8	36.43	50.46	---	---	71.61	55.70	57.57	**99.10**	98.72
	9	72.29	86.84	---	---	79.92	79.06	92.91	82.68	**99.50**
	10	4.92	1.83	---	---	81.78	63.97	**84.83**	53.08	69.86
	11	24.85	**75.34**	---	---	71.09	27.60	62.38	49.85	73.26
	12	28.07	62.41	---	---	10.39	58.26	0.25	**68.28**	57.27
	13	98.89	**99.17**	---	---	20.00	59.04	0	19.78	0
	14	88.02	86.34	---	---	78.93	55.08	91.05	96.24	**98.15**
	15	30.95	52.86	---	---	54.99	56.13	**62.10**	0	0
	16	14.14	5.94	---	---	72.14	16.17	0.46	97.64	**98.27**
User's accuracy (%)	1	48.54	55.67	OM	OM	0	62.99	45.29	39.96	**100**
	2	99.53	**99.72**	---	---	49.07	89.44	98.73	78.55	99.26
	3	33.70	35.83	---	---	45.91	55.51	80.15	55.63	**91.06**
	4	94.59	**94.74**	---	---	10.73	75.92	49.66	55.25	78.21
	5	88.45	**92.60**	---	---	53.04	70.34	64.19	66.11	66.26
	6	**100**	**100**	---	---	99.75	86.31	**100**	99.95	99.97
	7	88.88	90.49	---	---	96.02	69.00	77.87	98.85	**98.98**
	8	66.43	69.45	---	---	**73.83**	65.94	73.66	57.77	58.72
	9	64.80	79.83	---	---	92.33	92.43	92.07	82.12	**96.73**
	10	4.30	3.52	---	---	86.98	72.47	42.11	68.59	**90.18**
	11	19.22	32.50	---	---	**65.32**	33.93	59.64	17.32	63.55
	12	19.64	43.66	---	---	20.67	51.03	20.43	91.15	**94.77**
	13	**41.99**	30.85	---	---	5.65	31.23	0	18.53	0
	14	**76.20**	65.69	---	---	26.17	42.27	60.64	58.17	38.61
	15	42.87	53.04	---	---	**62.67**	37.30	53.01	0	0
	16	7.85	5.29	---	---	99.62	5.24	1.58	99.19	**99.83**
OA (%)		53.51	62.61	OM	OM	64.75	64.84	64.95	71.17	**77.43**
Kappa		0.4948	0.5894	OM	OM	0.6090	0.6101	0.6121	0.6741	**0.7455**
Purity		0.6660	0.6710	OM	OM	0.6917	0.6903	0.6860	0.7197	**0.7786**
Time (s)		1015	3011	OM	OM	24	61	**13**	2699	96241

1: Brocoli-green-weeds-1; 2: Brocoli-green-weeds-2; 3: Fallow; 4: Fallow-rough-plow; 5: Fallow-smooth; 6: Stubble; 7: Celery; 8: Grapes-untrained; 9: Soil-vinyard-develop; 10: Corn-senesced-green-weeds; 11: Lettuce-romaine-4wk; 12: Lettuce-romaine-5wk; 13: Lettuce-romaine-6wk; 14: Lettuce-romaine-7wk; 15: Vinyard-untrained; 16: Vinyard-vertical-trellis.

TABLE V
ANALYSIS FOR PARAMETER *s*

The size of the matching window *s*	2	3	5
Indian Pines image	63.45	63.27	63.27
Salinas image	77.43	77.43	77.43

D. Parameter Analysis

The proposed NL-SSLR algorithm totally includes six key parameters. The sensitivity of each parameter was analyzed in detail on both experimental data sets, with the other parameters fixed at their optimal values. Figs. 10–14 and Table V show the change trend of the OA of the proposed NL-SSLR algorithm versus each parameter on both experimental images. To better verify the performance of NL-SSLR, the optimal values of the clustering precisions of the other three state-of-the-art scalable methods are also included.

1) Parameter n: Parameter n is a size parameter for the dictionary, which is utilized to control the number of atoms in the sketched dictionary. It determines the quality of the dictionary, and it trades off the representative ability and the redundancy of the dictionary. As a result, n directly affects the

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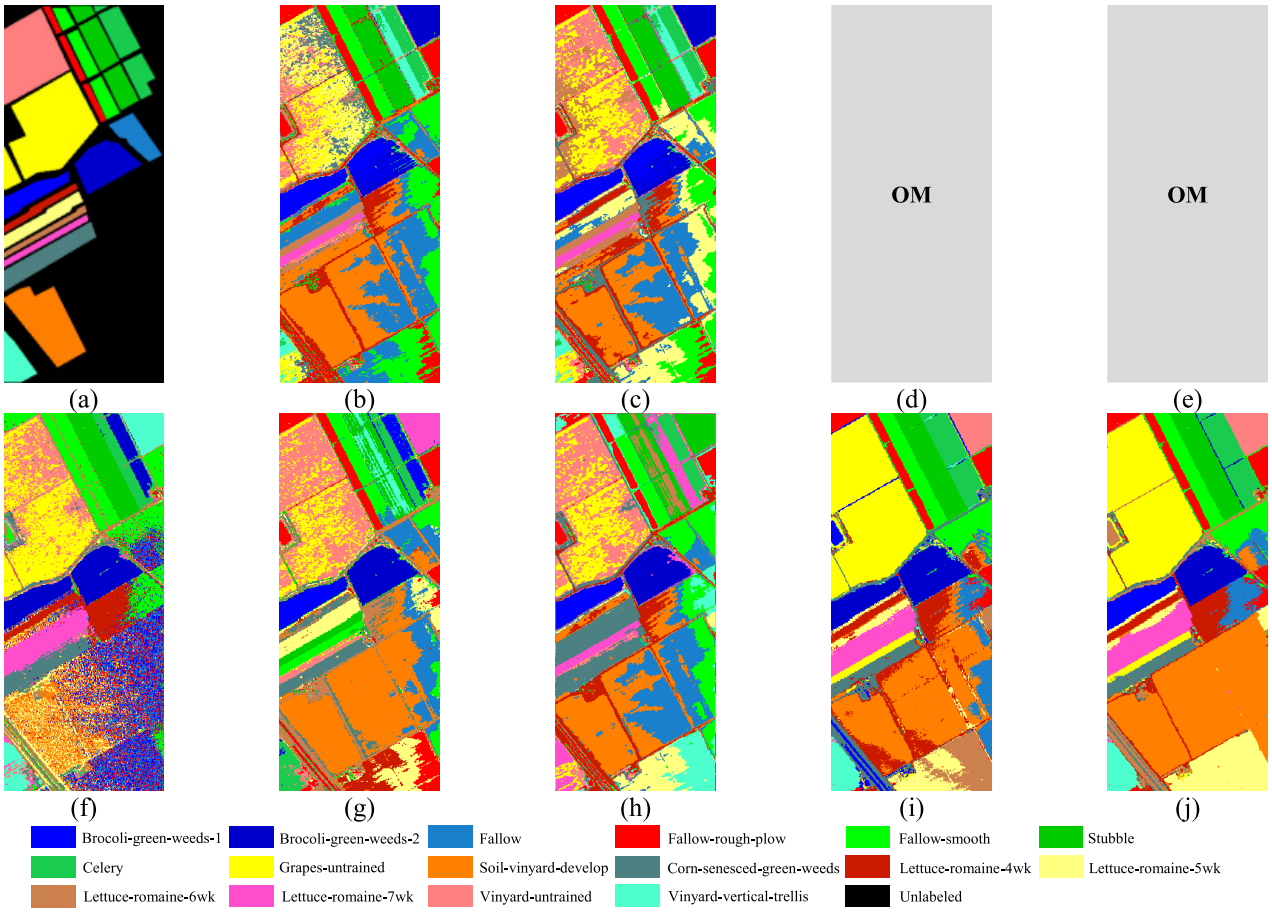


Fig. 9. Cluster maps of the Salinas experiment. (a) Ground truth. (b) FCM. (c) FCM-S1. (d) SSC (OM means out of memory). (e) L2-SSC (OM means out of memory). (f) Scalable-SC. (g) SGCNR. (h) FSCAG. (i) Sketched-SSC. (j) NL-SSLR.

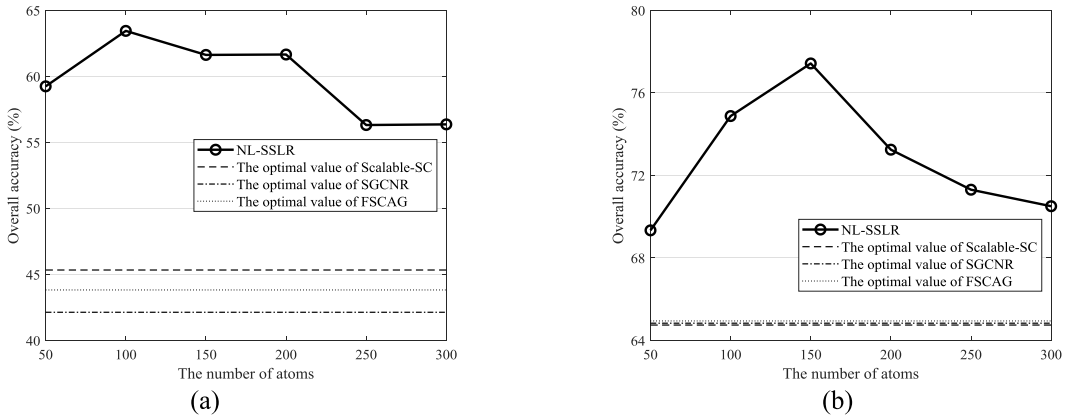


Fig. 10. Analysis for parameter n. (a) Indian Pines image. (b) Salinas image.

final clustering accuracy of NL-SSLR. In practice, parameter n obtains its value from the range of [50, 300], with an interval of 50.

As shown in Fig. 10, it can be found that the NL-SSLR algorithm is relatively robust to parameter n. The clustering performance of NL-SSLR is relatively stable for both scenes within a wide range of parameter n. In addition, NL-SSLR obtains much better performance than the other three scalable clustering methods, in all cases. In general, parameter n gets its optimal value from [100, 200] and can be easily determined in practice.

2) Parameter λ: Parameter λ is a regularization parameter used to tradeoff the representation accuracy and prior constraints and is generally decided by γ, as in SSC [12], [35]. Therefore, we only need to analyze the sensitivity of parameter γ in practice. Parameter γ gets its value from the range of [200, 1800], with an interval of 200 [12], [38], [39].

Through carefully observing Fig. 11, we can find that γ has relatively less influence on the final clustering accuracy of NL-SSLR. NL-SSLR can obtain a relatively high clustering accuracy in most cases, and the optimal value of γ is around 800 for both images. Similarly, NL-SSLR outperforms the

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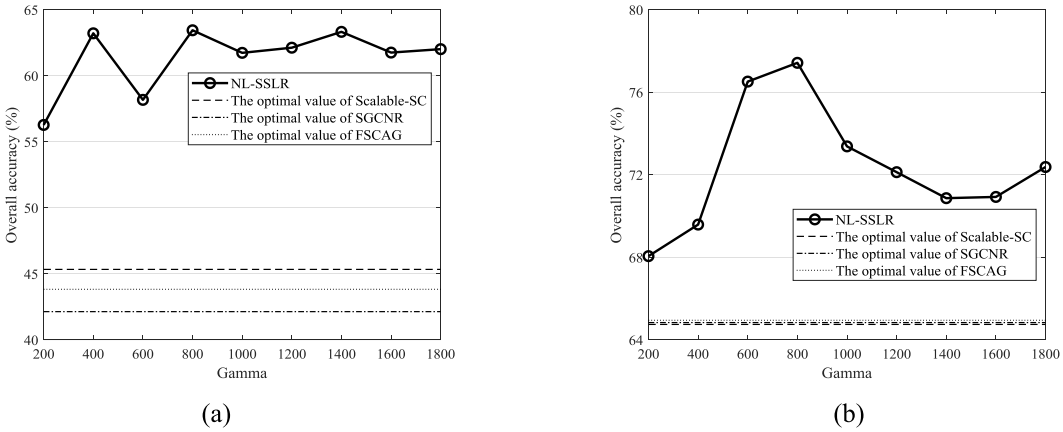


Fig. 11. Analysis for parameter γ. (a) Indian Pines image. (b) Salinas image.

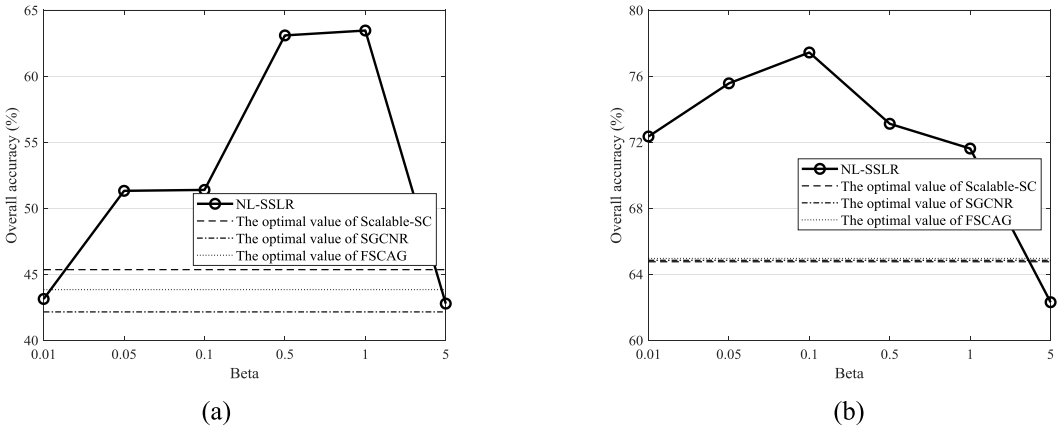


Fig. 12. Analysis for parameter β. (a) Indian Pines image. (b) Salinas image.

other three scalable methods, in all cases. In practice, this parameter can be easily fine-tuned.

3) Parameter β: Parameter β is utilized to balance the sparse property and low-rank property of the representation coefficient matrix. Through parameter β, we can effectively adjust the ratio of the local structural information and global structural information. In practice, it gets its value from the set of {0.01, 0.05, 0.1, 0.5, 1, 5}.

Fig. 12 shows that the parameter β has great effects on the final clustering accuracy of NL-SSLR. In general, with the change of β, NL-SSLR performs good in most cases, and it can obtain a higher accuracy than the other three scalable methods. Parameter β generally gets its optimal value from [0.1, 1] and can be easily determined in practice.

4) Parameter α: Parameter α is utilized to tradeoff the importance of the spectral information and spatial correlation information. It controls the strength of the nonlocal correlation information incorporated into the SC model and directly affects the final clustering accuracy. In practice, parameter α gets its value from the set of {0.01, 0.05, 0.1, 0.5, 1, 3, 5}.

In Fig. 13, it can be noted that the parameter α is also very important for the proposed NL-SSLR algorithm. However, generally speaking, the performance of NL-SSLR is relatively stable over a large range of α, and it obtains higher clustering accuracy compared with the other three scalable methods in all cases. This parameter is also easily fine-tuned in practice.

5) Parameter S: Parameter S is utilized to control the size of the search window. It determines the search range of the spatial correlation information. In addition, with the increment of S, the search time rapidly increases. In practice, parameter S gets its value from the range of [3, 13], with the interval being 2.

In Fig. 14, it can be found that NL-SSLR is relatively stable for various values of S, and it achieves a much higher clustering accuracy in all cases, compared with the other three scalable methods. Generally speaking, the optimal value of S can be easily determined in practice.

6) Parameter s: Parameter s is utilized to control the size of the matching window. It is generally related to the spatial homogeneity of the distribution of the land objects, and it determines the size of the spatial neighborhood. In practice, parameter s gets its value from the set of {2, 3, 5}.

As shown in Table V, a conclusion can be drawn that the influence of this parameter on the clustering accuracy of NL-SSLR is very small, and NL-SSLR can generally obtain a good performance with various s.

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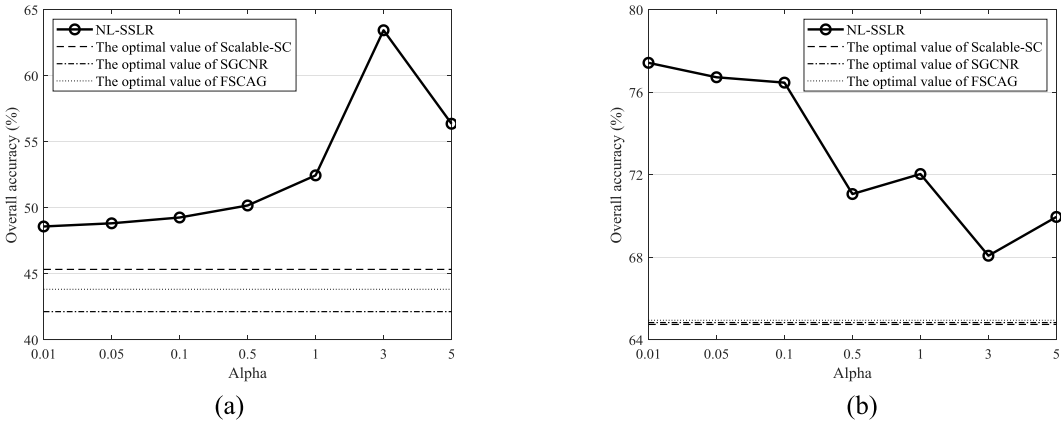


Fig. 13. Analysis for parameter α. (a) Indian Pines image. (b) Salinas image.

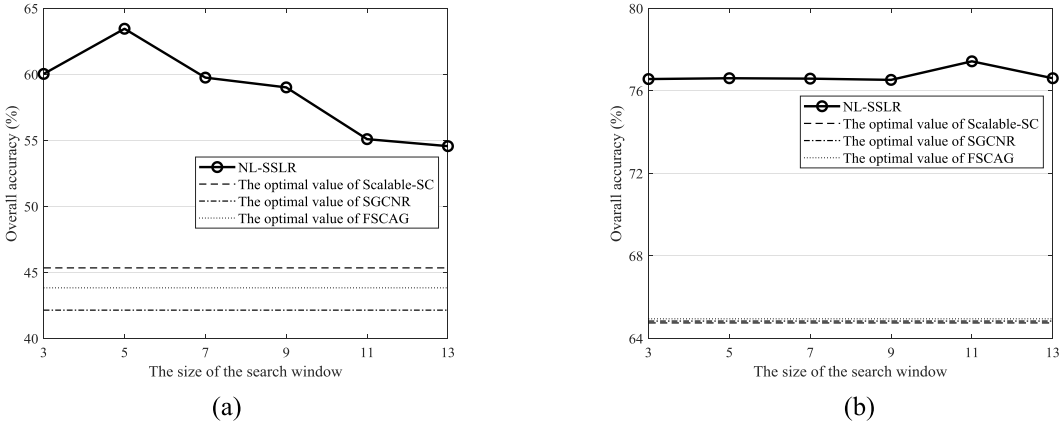


Fig. 14. Analysis for parameter S. (a) Indian Pines image. (b) Salinas image.

V. Conclusion

In this article, the recently developed sketched-SC model is introduced to solve the large HSI clustering problem. However, due to the inadequate mining of the structural information and the absence of spatial information, directly applying sketched-SC to HSIs generally fails to obtain a good performance. To overcome these obstacles, a novel scalable NL-SSLR model is presented for use with large HSIs. Therein, the underlying local and global structural information of the HSI can be simultaneously explored by the reweighted sparse and low-rank representation model. In addition, the spatial homogeneity of the clustering results can be better promoted by the nonlocal means regularization. NL-SSLR inherits the advantages of SC, while effectively improving its scalability, which makes it feasible for use with large HSIs. Several experiments demonstrated the effectiveness and superiority of NL-SSLR, and it can be an effective clustering tool for large HSIs.

However, the proposed NL-SSLR algorithm still has some limitations. For example, if the spatial correlation information could be modeled in a more efficient way, to reduce the time cost, the NL-SSLR model would be more attractive in practical applications. In addition, if NL-SSLR could be extended to a nonlinear model to better deal with the complexity of HSIs, the clustering accuracy would be further improved. On the other hand, if multiple features from different domains of HSIs could be comprehensively taken into account by NL-SSLR, the discriminant ability of the model would be effectively improved. Furthermore, adaptive parameter adjustment would further improve the practicability of the method. All of these will be the directions of the next efforts.

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